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Probing the *Qi* of traditional Chinese herbal medicines by the biological synthesis of nano-Au†

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Herbal medicines with different *Qi* properties (the primary proxy of their therapeutic effects) are used in traditional Chinese medicine to maintain the harmony of vital forces in a human body. In the Western medicinal practice, the classification of *Qi* into four major families (“*Si Qi*” in *Chinese Pinyin*) is a challenging endeavor, especially by a simple non-reductionist approach. The method presented here is however able to distinguish the *Qi* of herbal medicines based on the measurements of several *Qi*-related features in a biological synthesis of nano-Au in herbal extracts: solution color, surface plasmon resonance properties, reaction time and nano-Au morphology. These *Qi*-related features on their own do not form sufficiently distinct clusters that are useful for the classification of the *Qi*-properties. The power of differentiation, however, is significantly improved when all the *Qi*-related features are considered together. The statistics of differentiation is encouraging, enabling us to develop a scheme which can classify all of the tested TCHMs into their respective *Qi* families. While this classification method was developed using a limited number of herbal medicines with known *Qi* properties, it has the potential to be applied as a scientific quick test to determine the *Qi* of new herbal medicines or herbal concoctions. It is our aspiration that this study can generate more interest in the development of non-reductionist approaches to modernize the understanding of TCHMs.

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1. Introduction

Traditional Chinese Medicine (TCM) is an alternative medical system rooted in Eastern philosophy which has developed its own unique theranostic methodologies.^{1,2} Although TCM has a complete system of classifying the functional states; its unique and seemingly inaccessible language has mystified its treatment philosophy.³ For example, TCM practitioners describe the physiological and pathological states of a human body in terms of *Yin* and *Yang*. TCM believes that the imbalance of *Yin* and *Yang* is the cause for disease. Consequently, disease prevention and treatment is all about restoring a body's *Yin* and *Yang* to the balanced state. TCM relies more on the use of Traditional Chinese Herbal Medicines (TCHMs) than surgical procedures to deliver treatments.^{1,2,4,5}

TCHMs are categorized, according to their therapeutic effect on the *Yin/Yang* balance, into two broad categories of *Yang* and *Yin*, which can be further divided into four large *Qi* families (*hot*, *warm*, *cool* and *cold* (Table 1); collectively known as “*Si Qi*” in Chinese Pinyin). The clinical application of TCHMs is

governed by their *Qi* properties. As a basic principle, the *Yang* TCHMs are the cure for *Yin* diseases (*e.g.*, typhoid and diarrhea), while the *Yin* TCHMs are the cure for *Yang* diseases (*e.g.*, pyrexia and anorexia).^{4,5} It is also believed that the *Yang* medicines enhance the metabolism of organisms, while the *Yin* medicines do exactly the opposite by suppressing metabolism. Time-tested practices in medication are the basis of these principles.

Despite the accumulation of an immense amount of clinical knowledge about the *Qi* properties of TCHMs, the modern pharmacology approach to understanding the TCHMs (*i.e.*, identification of active ingredients, the pharmacokinetics and pharmacodynamics of TCHMs) is a relatively new development which began in the early 20th century.^{6,7} Current pharmacology investigations of TCHMs are conducted mainly using either the chemical approach or the biological approach.^{2,8} In the former, the active ingredients of a TCHM are isolated, chemically assayed, and catalogued. Given the extraordinarily large number of components in a TCHM, the identification of biologically active molecules responsible for the *Qi* properties by this reductionist approach is an extremely daunting, if not impossible, undertaking. The biological approach examines the therapeutic effects of TCHMs in animal studies, which involves not only an equally onerous effort but also ethical considerations. Although the reductionist approach has accomplished great success in Western medical research, the development of TCM and TCHMs

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Table 1 The TCHMs selected for the biological synthesis in this study, classified according to their “*Si Qi*” families with the arrow showing the increase in their *Yin* character

<i>Yin/Yang</i> Classification	“ <i>Si Qi</i> ” Families	Clinical Property	Selected TCHMs*
	Hot	Cure for <i>Yin</i> diseases (e.g., typhoid and diarrhea)	<i>Gan Jiang</i> (a1), <i>La Jiao</i> (a2), <i>Bi Ba</i> (a3), <i>Rou Gui</i> (a4)
	Warm		<i>Gui Zhi</i> (b1), <i>Ai Ye</i> (b2), <i>Cang Shu</i> (b3), <i>Rou Cong Rong</i> (b4)
	Cool	Cure for <i>Yang</i> diseases (e.g., pyrexia and anorexia)	<i>Tu Da Huang</i> (c1), <i>Bo He</i> (c2), <i>Xi Yang Shen</i> (c3), <i>Ly Cha</i> (c4)
	Cold		<i>Jin Yin Hua</i> (d1), <i>Pu Gong Ying</i> (d2), <i>Ye Ju Hua</i> (d3), <i>Shi Hu</i> (d4)

* The Chinese Pinyin names are the preferred names for the TCHMs in this paper. A cross reference with their longer scientific names can be found in the Supplementary Information.

is historically experiential with a strong emphasis on biological holism, *i.e.*, the appreciation that a therapeutic outcome is determined by the fault tolerance, cooperative action, and self-regulation inherent in a complex system. Hence it is more constructive to regard *Qi* as the collective property of a TCHM, and to develop a non-reductionist approach for its determination.

In this report, we will demonstrate the use of a biological synthesis of nano-Au⁹ in TCHM extracts to distinguish the TCHM *Qi* properties. The synthesis features such as solution color, reaction time; and the surface plasmon resonance (SPR) properties and morphology of nano-Au, were all found to be *Qi*-related. Through the statistical analysis of these *Qi*-related features, we formulated a classification scheme which is also useful as a quick scientific test to determine the *Qi* of new herbal medicines or herbal concoctions with unidentified *Qi* properties.

2. Experimental section

2.1. Biomass extraction of the TCHMs

The following procedure was used to emulate the brewing process of traditional Chinese medicines. 6 g of the selected TCHM was cut into small pieces and then boiled in 30 mL ultrapure water for 5 min to extract the biomass. The solid residue was removed by centrifugation at 10 000 rpm for 10 min. The supernatant was used as the TCHM extract stock solution for subsequent biosynthesis.

2.2. Biological synthesis of nano-Au

The stock solution of the TCHM extract was first diluted 10-fold. 100 μL of 10 mg mL^{-1} HAuCl_4 was added to the diluted TCHM extract and reacted at room temperature for 5 hours, which was the time long enough to complete all biological synthesis of nano-Au in this study. The reaction mixture at the completion of the reaction was centrifuged twice at 10 000 rpm for 10 min each. The supernatant was discarded, and 500 μL ultrapure water was added to the residue. The resulting mixture was centrifuged at 10 000 rpm for 10 min again to obtain the final nano-Au product.

2.3. Characterization

Transmission electron microscopy (TEM), high-resolution TEM (HRTEM), selected area electron diffraction (SAED), and

energy-dispersive X-ray spectroscopy (EDS) images were obtained with a JEM-2100F (JEOL, Japan) microscope operating at 200 kV accelerating voltage. Samples for TEM analysis were prepared by dispensing a drop of the nano-Au solution on an amorphous carbon-coated TEM copper grid. X-ray photoelectron spectroscopy (XPS) analysis was carried out on a Kratos Axis Ultra DLD spectrometer. All binding energies were corrected by referencing the C 1s peak of adventitious carbon to 284.5 eV. The phase purity of the biosynthesized nano-Au was assayed by X-ray diffraction (XRD) on a Bruker D8 advance X-ray diffractometer using a Cu K α radiation source (1.5405 Å). UV-vis absorption spectra of the reaction mixture were recorded by a Shimadzu UV-1800 spectrometer as a function of biosynthesis time. TCHM extracts without the gold precursor were used as the solution reference to minimize the interference from the TCHM extract color in UV/vis measurements.

2.4. Statistical analysis and calculations

See the supplementary text in the ESI.†

3. Results and discussion

3.1 Biological synthesis of nano-Au in TCHM extracts

In a typical biological synthesis, the biomass or its extract provides a broad spectrum of reducing and surface-active agents for forming noble metal nanoparticles from their precursor salts. The biosynthesized noble metal nanoparticles often display shapes and sizes not commonly found in a chemical synthesis with small molecules.¹⁰ Many types of biomass (*e.g.*, viruses, bacteria, fungi, and plants) and their extracts can be used for the synthesis.^{9–20} Since TCHMs are almost exclusively applied as extracts of their biomass,^{8,21,22} the *Qi* properties of the TCHM extracts which are used clinically for disease treatment may also contain the active ingredients for the synthesis of noble metal nanoparticles. Au nanomaterials were chosen as the biosynthesis target because of their size and shape dependent properties and their good chemical stability.²³ The ability to synthesize nano-Au in different TCHM extracts was the first test of the generality of this approach. After a preliminary screening, we selected the TCHMs in Table 1 (a detailed list of the chemicals, materials and

methods can be found in the ESI†) to represent the four major *Qi* families (*i.e.*, hot, warm, cool and cold). The selection was based on the consultation of two authoritative classical texts of TCM (*i.e.* *Shennong's Classic of Materia Medica* and *Compendium of Materia Medica*) and the confirmation of the representativeness of the “*Si Qi*” properties by TCM practitioners.

The biosynthesis of nano-Au was based on the procedures from our previous work^{16,17} with some slight modifications. Briefly, a TCHM was boiled in ultrapure water for 5 min to form a herbal extract. The solid residue was removed by centrifugation. These steps were used to emulate the TCHM brewing process. The as-prepared TCHM extract was then reacted with a gold precursor (HAuCl₄) solution to form the nano-Au. The reaction was allowed to run to completion at room temperature. An illustration of the complete workflow is shown in Fig. 1a. The biosynthesis progress was followed by measuring the changes of the synthesis solution color and the intensity of the SPR peak of nano-Au with time by ultraviolet-visible (UV-vis) spectroscopy. The shape and size of the final nano-Au products and their distributions were examined by TEM. The TCHM *Bi Ba* from the *hot* family will be used as an example in the following discussion.

The HAuCl₄ Au precursor solution was added to the *Bi Ba* extract without any other reducing and stabilization agent (see Materials section in the ESI†). Reduction was followed in real-time by recording the intensity changes of the nano-Au SPR peak at 550 nm. Fig. 1b shows the evolution in the UV-vis spectra of the biosynthesis solution over a total reaction time of ~200 min (as determined from the SPR peak intensity–time plot in Fig. S1, ESI†). The color of the final nano-Au solution was pink (Fig. S1, ESI†), a warm color (color temperature of 2700–3000 K).²⁴ TEM examination (Fig. 1c) revealed the nano-Au product as a mixture of nanoparticles (NPs, ~10 to 50 nm) and nanoflakes (NFs, ~100 nm in lateral size). The effect of the extract concentration on biosynthesis was also investigated; where repeated measurements showed no significant changes in the nano-Au shape and size (Fig. S2, ESI†). A HRTEM image (Fig. 1d) of a typical Au NP and its corresponding SAED pattern (Fig. 1e) indicate that the nano-Au was crystalline. The same also applied to the NFs. The existence of zero-valent Au was confirmed by XRD (Fig. 1f) and XPS (Fig. 1g). XPS and EDS also detected carbon, nitrogen and oxygen; an indication of the presence of biomolecular residues on the nano-Au (Fig. S3 and S4, ESI†).

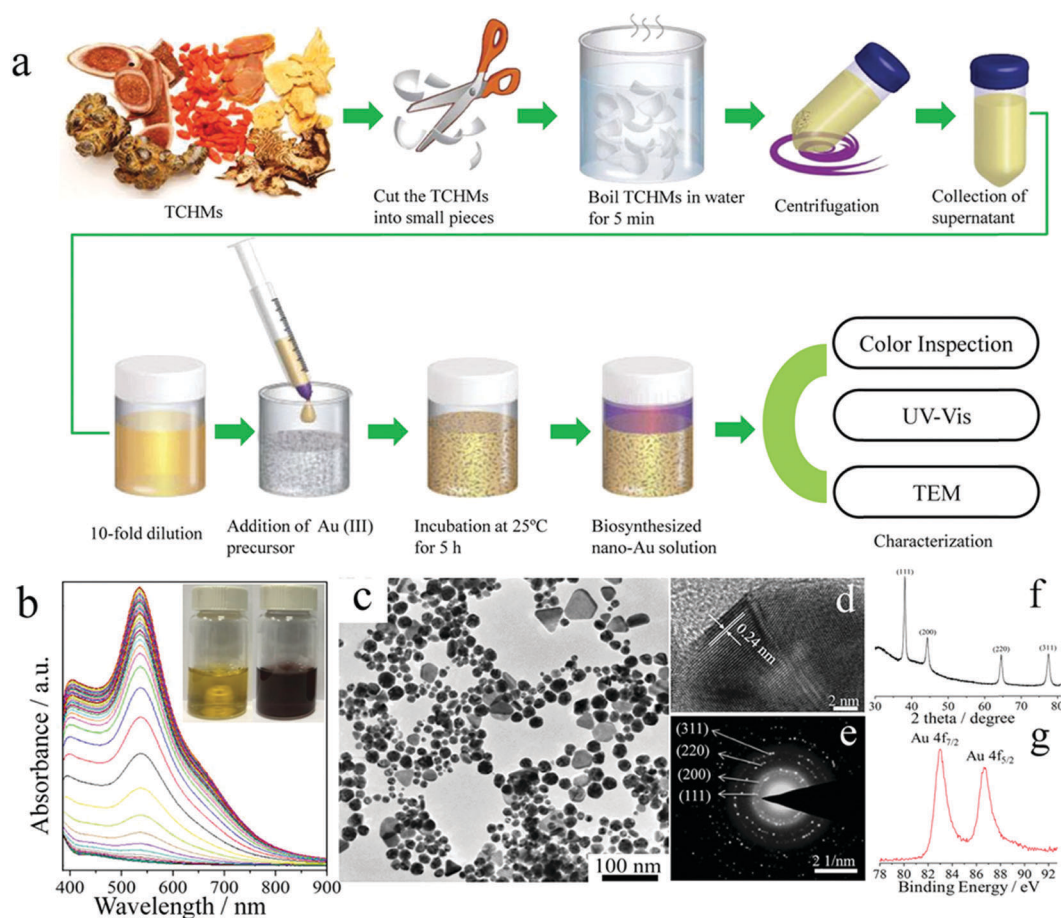


Fig. 1 Biological synthesis of nano-Au from raw TCHMs. (a) Schematic illustration of the steps in the preparation of a TCHM extract, the biological synthesis of nano-Au, and the measurement methods. Characterization of nano-Au synthesized by the *Bi Ba* extract. (b) Time-resolved UV-vis spectra. Insets: Digital photos of the *Bi Ba* extract (left) and the nano-Au solution at the completion of the biosynthesis (right). (c) TEM image of the nano-Au products and (d) HRTEM image of a representative Au NP with its (e) SAED patterns, (f) XRD pattern, and (g) high resolution XPS spectrum in the Au 4f region.

Table 2 Au nanomaterials synthesized by TCHMs with different *Qi* properties in terms of solution color, characteristics of the Au SPR peak, reaction time, and nano-Au morphology (shape and size)

Yin or Yang	“Si Qi”	TCHMs	Solution color	UV-vis measurements		TEM measurements (shape & size)
				SPR peak intensity	Reaction time (min)	
Yang	<i>Hot</i>	a1	Pink	Increasing	~120	NPs: 10–50 nm & NFs: ~100 nm
	<i>Hot</i>	a2	Pink	Increasing	~250	NPs: 10–50 nm & NFs: ~100 nm
	<i>Hot</i>	a3	Pink	Increasing	~200	NPs: 10–50 nm & NFs: ~100 nm
	<i>Hot</i>	a4	Pink	Increasing	~100	NPs: 10–50 nm & NFs: ~100 nm
	<i>Warm</i>	b1	Pink	Increasing	~30	NPs: 10–50 nm
	<i>Warm</i>	b2	Gray-blue	Increasing	~50	SA-NSSs: 100–500 nm
	<i>Warm</i>	b3	Pink	Increasing	~30	NPs: 10–50 nm
	<i>Warm</i>	b4	Pink	Increasing	~30	NPs: 10–50 nm & NFs: ~100 nm
Yin	<i>Cool</i>	c1	Gray-blue	Increasing	~30	NPs: 10–50 nm & SA-NSSs: 100 nm
	<i>Cool</i>	c2	Gray	Decreasing	~60	SA-NSSs: 100–500 nm
	<i>Cool</i>	c3	Pink	Increasing	~200	NPs: 10–50 nm
	<i>Cool</i>	c4	Blue	Increasing	~5	SA-NSSs: 100–500 nm
	<i>Cold</i>	d1	Gray-blue	Decreasing	~30	SA-NSSs: 50–500 nm
	<i>Cold</i>	d2	Gray-blue	Decreasing	~30	SA-NSSs: 50–500 nm
	<i>Cold</i>	d3	Gray-blue	Increasing	~30	SA-NSSs: 50–500 nm
	<i>Cold</i>	d4	Blue	Increasing	~60	SA-NFs: ~100 nm

NP: nanoparticle. NF: nanoflake. SA-NSSs: self-assembled nanospheres. SA-NFs: self-assembled nanoflakes.

The same biosynthesis protocol was applied to the TCHMs in Table 1. The resulting nano-Au products were characterized in the same way as in the *Bi Ba* example. Table 2 summarizes the results in terms of the four *Qi*-related features (*i.e.*, solution color, SPR intensity, reaction time and nano-Au morphology) of biosynthesis with these representative TCHMs.

3.2. Identifying the *Qi*-related features in the biosynthesis

Different from studies on the chemical and biological synthesis of nano-Au where the objective is to obtain uniformity of size, shape or self-assembled nanostructures by the optimization of the synthesis conditions,^{14,25–28} the biosynthesis in this study was carried out under a common set of conditions; using the

Qi properties of the TCHMs to generate a diversity of shapes, sizes and assembled nanostructures; which were then analyzed statistically together with other *Qi*-related synthesis features (*i.e.*, solution color, SPR properties, and reaction time) to generate recognition patterns for the classification of the TC HM *Qi* properties.

Some causality between the biosynthesis features and TCHM *Qi*-properties was revealed by comparing the observations from TCHMs in the same *Qi* family and between different *Qi* families. The results from the *hot* and *cold* families were compared first since these two families were the furthest apart in *Qi* properties (Table 1). Fig. 2a shows that the nano-Au synthesized with *Rou Gui* (**a4**, from the *hot* family) was a mixture of discrete NPs

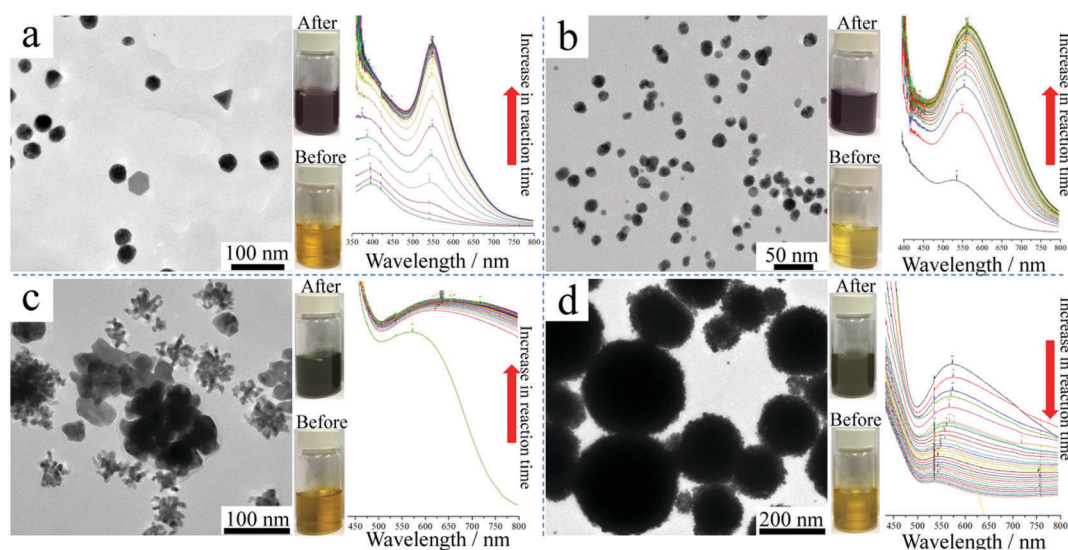


Fig. 2 The typical nano-Au products synthesized by TCHMs from different *Qi* families. (a) *Rou Gui* (*hot* family), (b) *Gui Zhi* (*warm* family), (c) *Tu Da Huang* (*cool* family), and (d) *Jin Yin Hua* (*cold* family) with their corresponding TEM images (left), digital photos of the nano-Au solution before and after the biosynthesis (middle), and time-resolved UV-vis spectra (right).

(~10–50 nm) and NFs (~100 nm); and the nano-Au solution displayed a warm color. All TCHMs from the *hot* family also showed the following common characteristics: a long bio-synthesis completion time (>60 min) and an increasing SPR peak intensity with reaction time (Table 2 and Fig. S5, ESI†). The results from *Jin Yin Hua* (d1) and other members of the *cold* family are diametrically different (Fig. 2d): there were no discrete NPs or NFs; the nano-Au products were all self-assembled supra-nanostructures (denoted as SA-SNSs) including self-assembled nanospheres (SA-SNs, ~100–500 nm) and self-assembled nanoflakes (SA-NFs, ~100 nm) (Table 2 and Fig. S6, ESI†). The nano-Au solutions all had a cool color (e.g., the blue color of the nano-Au solution synthesized with *Jin Yin Hua* (Fig. 2d)). Biosynthesis with a *cold* TCHM could also be completed in less than 60 min.

The biosynthesis features are therefore categorically similar for TCHMs from the same *Qi* family; and distinctively different for TCHMs from different *Qi* families. TCHMs from the same *Qi* family produced the same solution color attribution (warm color from the *hot* family and cool color from the *cold* family), the same type of nanostructures (discrete nanostructures from the *hot* family and self-assembled supra-nanostructures from the *cold* family), and the same zone of reaction time (>60 min from the *hot* family and <60 min from the *cold* family).

Such contrasting observations were however not found between the *warm* and the *cool* families. In contrast, there were instances where TCHMs from the same *Qi* family would produce an “out-of-category” observation common to the TCHMs from other families (Table 2 and Fig. S7, S8, ESI†). For example, SA-SNs, which were found previously only in biosynthesis with TCHMs from the *cold* family (Fig. S6, ESI†), were also present among the nano-Au products synthesized by the extracts of *Ai Ye* (b2, from the *warm* family) and *Bo He* (c2, from the *cool* family). The *warm* and *cool* families therefore do not form morphologically distinctively different nano-Au products. This could be due to the lack of a sufficiently large *Qi* property difference between the *warm* and *cool* families (Table 1). The lack of mutually exclusive nano-Au morphologies also occurred between other *Qi* families (Fig. 3). For example, exclusively discrete

nanostructures which were hitherto unique only to the *hot* TCHMs could also be formed by some TCHMs from the *warm* family (i.e., overlap-1). The overlap of nano-Au morphologies was however greater between the *warm* and *cool* families (overlap-2); and between the *cool* and *cold* families (overlap-3). As a result, the “NPs only” and the “SA-SNs only” morphologies could be found in both *warm* and *cool* TCHMs with a probability of 50%. Even for the two nonadjacent *Qi* families of *warm* and *cold*, the SA-SNs morphology was found to be an overlapped *Qi*-related feature (overlap-4).

Despite the overlap of the nano-Au features, the nano-Au morphology shifted progressively from discrete nanostructures to self-assembled supra-nanostructures with the change of the TCHM from the *hot* family to the *cold* family. There were also overlaps in other *Qi*-related features besides the nano-Au morphology (Table S1 and Fig. S9, ESI†). All of these observations indicate that individual *Qi*-related features do not provide distinct boundaries which can be used to distinguish between TCHMs from different *Qi* families. Therefore, the classification of TCHMs into their respective *Qi* families cannot be achieved by analyzing the four *Qi*-related features in isolation.

3.3. Developing a classification scheme for the *Qi* of TCHMs

We then considered all *Qi*-related features together to improve the power of discrimination between the *Qi* families. To do so we calculated the discrimination probabilities of all *Qi*-related features based on the frequencies of successful discriminations between two different *Qi* families, which are summarized in Table 3 (with details of the calculations in the supplementary text in the ESI†).

The discrimination probability of a *Qi*-related feature is the probability of differentiating any two *Qi* families by that *Qi*-related feature alone. A high discrimination probability is obtained when the *Qi*-related features from different *Qi* families are contrasting enough and there is minimal occurrence of common observations. For example, the *hot* and *cold* families can be completely distinguished by using the solution color attribution (warm or cool color); and hence $P_{\text{color}} = 1$ for the *hot/cold* differentiation. In contrast the probability of successfully distinguishing between *hot* and *warm* families by the solution color attribution is only 25% ($P_{\text{color}} = 1/4$ for the *hot/warm* differentiation). While some *Qi* families can be distinguished quite confidently based on the measurements of only one *Qi*-related feature (e.g., $P_{\text{TEM}} = 1$ between the *hot* and *cold* families based on the presence or absence of mutually exclusive discrete and self-assembled supra-nanostructures), this is more

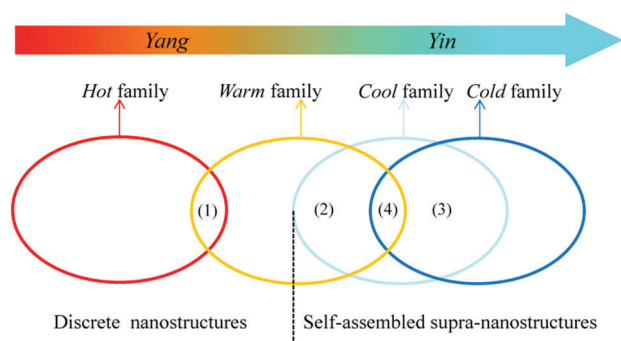


Fig. 3 Schematic illustration showing the overlaps of nano-Au morphology in biosynthesis with TCHMs of different *Qi* properties. While there is no overlap between the *hot* and *cool* or *cold* families; there are overlaps between the *hot/warm* (1), *warm/cool* (2), *cool/cold* (3) and *warm/cold* (4) families.

Table 3 Discrimination probabilities of each *Qi*-related feature between two different TCHM families

“Si Qi” comparison	P_{color}	P_{SPR}	P_{Time}	P_{TEM}
<i>hot/warm</i>	1/4	0	1	3/4
<i>hot/cool</i>	3/4	1/4	3/4	1
<i>hot/cold</i>	1	1/2	15/16	1
<i>warm/cool</i>	5/8	1/4	1/2	3/4
<i>warm/cold</i>	3/4	1/2	1/4	3/4
<i>cool/cold</i>	1/4	1/2	9/16	1/2

of an exception than the norm. The discrimination probability can however be improved by considering all four Q_i -related features (solution color, SPR intensity, reaction time and nano-Au morphology) together. A hierarchy of tests for identifying the Q_i family was developed after a detailed statistical analysis of the Q_i -related features. Since the determination of solution color attribution is the simplest measurement which can be done without any equipment, the solution color attribution was chosen as the first classification test. Among the four Q_i -related features, the nano-Au morphology contains the most information and distinguishability (*i.e.* as mixtures of NPs and NFs, NPs only, mixtures of NPs and SA-NSs, and SA-SNSs only); and hence has the highest discriminative power; as shown by the P_{TEM} being all greater than 1/2 in Table 3. It is however the costliest test to run (in terms of time, effort, and infrastructure); and was therefore relegated to the last test. The measurements of SPR intensity and reaction time are both based on the UV-vis spectroscopy. It was found that reaction time (sorted into three zones, *i.e.*, $t < 1$ h, $1 \text{ h} \leq t < 2$ h, and $t \geq 2$ h) has a greater discriminative power than the SPR intensity (in terms of an increasing or a decreasing trend). Consequently, reaction time and SPR properties were assigned respectively as the second and third tests in the classification scheme.

Fig. 4 shows the probability-based classification scheme using all four Q_i -related biosynthesis features (*i.e.*, color, SPR peak intensity, reaction time, and nano-Au morphology) to deduce the Q_i family of a TCHM. This is able to distinguish all TCHMs from known Q_i families in this study. This also has the potential to determine the Q_i category of new TCHMs or

TCHM concoctions. The following is the description of the steps in the application of the scheme.

(1) The color attribution of the nano-Au solution (warm or cool) can be used to sort the TCHMs into two distinct groups: a warm color (*e.g.*, pink) for the *hot*, *warm* and *cool* families; and a cool color (*e.g.*, blue) for the *warm*, *cool* and *cold* families. There are multiple Q_i families in each group due to the overlap of this Q_i -related feature.

(2) The SPR peak intensity (in increasing or decreasing trend) is then applied as the next test of classification. *E.g.*, in the overlapped *hot*, *warm* and *cool* families, a decreasing SPR peak intensity with reaction time can only be caused by the *cool* family. On the other hand, the overlapped *warm*, *cool* and *cold* families do not provide a clear-cut answer for this test – a decreasing SPR intensity with time is nonetheless useful to reduce the overlap to two Q_i families: *cool* or *cold*. An increasing SPR intensity with time offers no useful information for classification, and additional Q_i -related features (*i.e.*, reaction time and nano-Au morphology) must be used; as will be shown in the steps below.

(3) Reaction time divided into three different zones of $t < 1$ h, $1 \text{ h} \leq t < 2$ h, and $t \geq 2$ h is used as the next classification criterion. Fig. 4 shows that the overlap is further reduced after this step; where some of the results are exclusive to only the *cool* family.

(4) Finally, based on the four morphological classes of the nano-Au products as examined by TEM (*i.e.*, NPs & NFs, NPs only, NP & SA-NSs, and SA-SNSs only), the remaining Q_i families may be identified; all with probabilities greater than 50%.

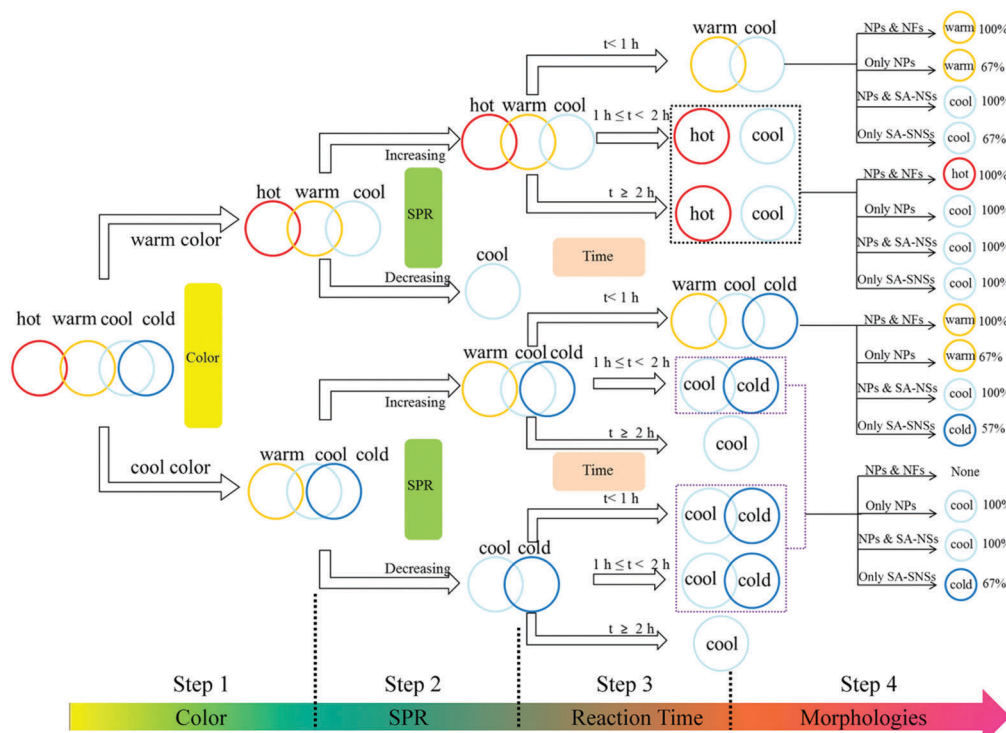


Fig. 4 A classification scheme for the Q_i -properties of herbal medicines based on the four Q_i -related features in the biosynthesis of nano-Au.

4. Conclusions

The feasibility of using the biological synthesis of nano-Au to categorize the *Qi* properties of TCHMs was demonstrated. The biosynthesis method as described here uses TCHMs in their native state; yielding several experimental features (e.g., solution color, SPR peak intensity, reaction time and nano-Au morphology) which are effective for determining their “*Si Qi*” family. The analysis of the experimental results revealed that the individual *Qi*-related features do not have sufficient discriminative power to identify the *Qi* family. However, a statistical analysis of the distribution of these *Qi*-related features enabled a probability-based classification scheme to be developed based on a hierarchy of decision-making tests. The contribution of this work is to demonstrate a non-reductionist scientific approach for the categorization of the TCHM *Qi* properties; which has the potential to be developed further for the determination of the *Qi* of new herbal medicines or herbal concoctions. As such it is expected to help accelerate TCM modernization and the development of TCHM standardization.

Conflicts of interest

There are no conflicts to declare.

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